



2310 'Encoder communication'

• Error in communication with encoder • Encoder does not report/no encoder present • Encoder cable defective or not connected • Error in encoder database									
Device									
Description									
Class	Warning								
Drive Behaviour	Drive runs down								
Device Behaviour	System run-up aborted								
Additional Error Information (AMK Service)									
	Error in communication:								
	Info 1	1	Timeout: No acknowledgement of the encoder						
		2	CRC error: Faulty transmission of the encoder						
		3	Alarm bit: The internal error bit of the encoder is set						
		4	No single or multi-turn encoder recognized.						
			Info 2	0	ID32776 'Sine encoder period' is higher than the real solution of the single turn encoder. Check parameter ID32953 'Encoder type'				
			P- / Q-encoder: ID34265 'Encoder ratio'						
			Info 2	1	Motor revolution or encoder revolution not equal 1				
		2		No power of two					
		3		Motor revolution not equal 1 on singleturn encoder					
		5	Handshake: The answer of the encoder is wrong, disturbed communication, damaged encoder.						
	6	Command cannot be executed by driver since encoder not yet initialized Help: Start System run-up again							
	7	Motor was in motion when reading the position							
		<table><tr><th>Trigger</th><th>Measure</th></tr><tr><td>AFP command READ_SINCOS</td><td>Increase ID57 'In position window'</td></tr><tr><td>Run-up</td><td>Motor must stand still during encoder run-up</td></tr></table>		Trigger	Measure	AFP command READ_SINCOS	Increase ID57 'In position window'	Run-up	Motor must stand still during encoder run-up
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8	Encoder property not compatibility, e.g. digital resolution to high, encoder type at present not possible.								
9	Signal error at the interpretation of the analog pulse (SIN, COS is not equal to the absolute pulse). Help: Check the cable of the encoder feedback.								
	Error in action with encoder database:								
	Info 1	10	Identification: Reading error ID32842 'Encoder list customer', the encoder data are invalid Help: The ID32842 must be recorded with valid data.						
		11	Checksum wrong: Data of ID32842 'Encoder list customer' in the encoder are wrong. Data of the were changed in the system without saving them in the encoder						
		12	ID32842 'Encoder list customer' was changed without saving the data in the encoder						
		13	Identification: Reading error "encoder list motor", the wrong list is invalid or the list is empty. Help: The encoder list motor must be recorded trough AMK with valid data.						

		14	Checksum wrong: Data of the “Motor encoder list” in the encoder are wrong. Data of the “Motor encoder list” were changed in the system without saving them in the encoder																																																																												
		15	The data of the ID No. listed in the “User encoder list” require more memory than available in the encoder																																																																												
		16	The addressed memory cell may not be written																																																																												
		17	Entered address does not exist																																																																												
		18	The memory range in the encoder cannot be determined correctly																																																																												
		19	The offset address in the encoder cannot be determined correctly																																																																												
		20	ID34160 'Part number motor' is not equal with the part number motor witch is written into the encoder Help: Read the motor data from the encoder.																																																																												
		21	In DSL-encoder is no database structure applied																																																																												
		33	Setpoint in ID32942 has a invalid polarity sign, referred to the motor sense of rotation ID32773, bit 16																																																																												
		34	The absolute setpoint in ID32942 is over the limit of the encoder (ID116 x distinguishable revolutions of the encoder)																																																																												
		70	A communication to FPGA has not been established																																																																												
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				optimal Info 3RSSI-register < 6, maximum 12	
			5	Error: Received signal strength is bad Info 3RSSI-register < 2	
			6	Warning: Signal quality is not optimal Info 3QM-register < 15	
			7	Error: Signal quality is bad Info 3QM-register < 8	
			8	Error: Connection to DSL-encoder was lost	
			9	Error: More than 6 positions was roughly because of errors Info 3ACCERR-register	
			10	Warning: In event register error bits was set Info 3EVENT-register	
			11	Error: POS bit set longer than allowed (Estimator active) Info 3Number of cycles until error message	
	Info 1	81	Error group= 40h DSL-encoder has been detected an error with a resource access Info 3error code		
		82	Error group = 41h DSL-encoder has been detected an error with a resource access Info 3error code		
		83	error group = 42h DSL-encoder has been detected an error with a resource access Info 3error code		
		84	Error group = 43h DSL-encoder has been detected an error with a resource access Info 3error code		
		85	Error during initialization of the DSL encoder		
		86			
	Errors in access to data stock:				
	Info 1	30	Reading access:Error on reading data from DTH		
		31	Writing access:Error on writing data into DTH		
		32	ID wrong: ID number not available		
		40	Inadmissible status		
		41	Timeout encoder commanding		
Note	Encoder errors of the option card KU-/KW-EN1 are distinguished from the motor encoder by Info 1 + 100				
Error Removal	• Check cable or plug • Check cable shield • Reduce cable length Info 1 = 17 Info2 = 5 or 6: Fault or defect of the controller card • Drive OFF / ON. Replace the control card if it recurs. Info1 = 80, 85, 86 Check encoder cable / POS bit is set on the DSL line due to faults / POS bit is set due to vibrations • Check the shielding of the DSL line • Check the mechanics for vibrations • Check ID33194 and ID33195				

Device	IDT					
Description	Messages during run-up (ID32773.23.= 1: reading the absolute position during initialisation)					
Class	see below					
Drive Behaviour						
Device Behaviour	System run-up aborted					
Additional Error Information (AMK Service)						
	ID32773.27 = 0					
	Info 1	6	Warning Monitor of the UPS supply has triggered			
		9	Error Actual position value change has been detected while no UPS was active (position reference has been lost).			
	ID32773.27 = 1					
	Info 1	3	Error			
			Info 2	10	Survey of valid Flash has activated	
		9	Error			
	Faulty encoder angle was detected during system run-up					
Device	IDT					
Description	Error in combination with absolute encoder (type C)					
Class	see below					
Drive Behaviour						
Device Behaviour						
Additional Error Information (AMK Service)						
	Info 1	3	Absolute encoder error			
			Info 2	1	Speed higher than 6000 rpm	
				2	Motor movement during the encoder power down (no battery power supply mode)	
				6	Communication error during the read access	
				7	Communication error during the write access	
				8	System SW error	
				9	Position lost because of the deep low power mode command	
				10	No valid record for position data is detected	
		5	Error Encoder communication error			
		6	Warning Absolute encoder power off detected			
		8	Error Faulty encoder firmware			
		9	Error Motor movement in low power mode without USV			
Error Removal	Encoder included in motor electric Message incl. info 1 and info 2 to AMK service for error localisation					

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